

Education Investment under China’s Hukou System: A Theoretical Framework

pAI-Econ-claude (theoretical-economics-claude-skill)

<https://github.com/maxwell12732/pAI-Econ-claude>

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Draft: June 14, 2026

1 Introduction

This paper develops a formal model of educational investment under China’s hukou (household registration) system. A rural child and an urban child of identical innate ability θ face different expected returns to education: the rural child earns a fraction $\delta \in (0, 1)$ of the urban wage unless she crosses an elite education threshold $\hat{e}$, at which point she gains probability $p_0 > 0$ of accessing the full urban labor market. We embed this two-regime payoff structure into the [Becker \(1964\)](#) human capital framework and characterize the resulting investment patterns.

Our central finding is that the hukou system generates a *non-monotone* relationship between ability and investment distortion. Low-ability rural children rationally under-invest relative to equally-able urban peers, confirming the benchmark of [Arrow \(1973\)](#): a proportional wage penalty suppresses the marginal return to education and reduces optimal investment. Intermediate-ability rural children, however, rationally *over-invest*. For these children, whose unconstrained optimum lies just below $\hat{e}$, the escape-route premium—the expected wage gain from crossing the elite threshold—outweighs the cost of over-shooting. The institutional barrier does not uniformly suppress investment; for those near enough to the threshold to find the gamble worthwhile, it reverses the direction of distortion.

Two further results follow. First, the welfare loss from hukou is non-monotone in ability, peaking among intermediate-ability children near $\hat{e}$ who bear the cost of over-investment with certainty while receiving the escape premium only with probability p_0 . Second, hukou reform that equalizes returns—as China’s post-2014 reforms aim to do—simultaneously improves welfare for all rural children and reduces educational attainment among the intermediate-ability group. The escape-route incentive that was driving over-investment disappears; these children now invest at their unconstrained optimum, which is strictly below $\hat{e}$. A policymaker monitoring Gaokao preparation intensity as a reform indicator would observe declining effort among the most motivated rural students precisely when the reform is working.

2 Model

2.1 Agents and Ability

Two children, an urban child C_U and a rural child C_R , share innate ability $\theta \in [\theta_L, \theta_H]$ with $\theta_L > 0$. Hukou registration status $\tau \in \{U, R\}$ is fixed at birth and determines labor-market access. Each child independently chooses education $e \in [0, \bar{e}]$ to maximize the present discounted value of lifetime earnings net of education costs, discounted at rate $r > 0$. Agents are risk-neutral.

Assumption 1 (Equal Ability). Both children possess ability θ ; the investment gap $\Delta e^*(\theta) := e_U^*(\theta) - e_R^*(\theta)$ is analyzed parametrically in θ to isolate the institutional effect of hukou from any ability differences between the urban and rural populations.

2.2 Wages and the Hukou Penalty

The market wage function is $w(e, \theta) = A\theta^\alpha e^\beta$, where $A > 0$, $\alpha > 0$, and $\beta \in (0, 1)$. The Cobb-Douglas specification implies strictly diminishing marginal returns to education ($w_{ee} < 0$) and strict supermodularity ($w_{e\theta} > 0$): higher ability raises the marginal return to education at every level e .

Assumption 2 (Multiplicative Penalty). The rural wage in the absence of an escape route is $\delta \cdot w(e, \theta)$, $\delta \in (0, 1)$. The penalty scales the entire return to education proportionally.

The escape-route probability function is

$$p(e) = \begin{cases} 0, & e < \hat{e}, \\ p_0, & e \geq \hat{e}, \end{cases} \quad (1)$$

where $\hat{e} \in (0, \bar{e})$ is the elite education threshold and $p_0 \in (0, 1)$. In the Chinese context, $\hat{e}$ corresponds to crossing the Gaokao provincial key-line for admission to a nationally recognized university, and p_0 captures the fraction of elite rural graduates who successfully access the urban labor market. The discontinuity at $\hat{e}$ reflects the binary nature of university-tier assignment.

2.3 Payoffs

Let $C(e)$ denote education costs with $C' > 0$ and $C'' > 0$. The urban child's payoff is

$$V_U(e; \theta) = \frac{w(e, \theta)}{r} - C(e). \quad (2)$$

The rural child faces a two-regime structure. Below the threshold,

$$V_R^{low}(e; \theta) = \frac{\delta w(e, \theta)}{r} - C(e), \quad e < \hat{e}. \quad (3)$$

At or above the threshold, the expected payoff reflects the escape-route lottery:

$$V_R^{high}(e; \theta) = \frac{[\delta + (1 - \delta)p_0]w(e, \theta)}{r} - C(e), \quad e \geq \hat{e}. \quad (4)$$

The effective wage multiplier $\delta + (1 - \delta)p_0 = 1 - (1 - \delta)(1 - p_0)$ lies strictly between δ and 1: crossing the elite threshold improves but does not fully equalize expected wages, because the escape succeeds only with probability $p_0 < 1$.

3 Equilibrium Characterization

Lemma 1 (Existence and Characterization). *Under the Cobb-Douglas wage and strictly convex cost:*

(i) *The urban child's optimum $e_U^*(\theta)$ uniquely solves*

$$\frac{\beta A \theta^\alpha (e_U^*)^{\beta-1}}{r} = C'(e_U^*). \quad (5)$$

Moreover, $e_U^(\theta)$ is strictly increasing in θ .*

(ii) *The rural child's global optimum satisfies*

$$e_R^*(\theta) \in \arg \max \left\{ \max_{e < \hat{e}} V_R^{low}(e; \theta), \max_{e \geq \hat{e}} V_R^{high}(e; \theta) \right\}.$$

The low-regime maximum is the unique interior point $e_{int}^(\theta)$ solving $\delta w_e(e_{int}^*; \theta)/r = C'(e_{int}^*)$. The high-regime maximum is at the interior point $e_{high}^*(\theta)$ solving $[\delta + (1 - \delta)p_0]w_e(e_{high}^*; \theta)/r = C'(e_{high}^*)$ when $e_{high}^*(\theta) \geq \hat{e}$, and at the corner $\hat{e}$ otherwise. Since the effective multiplier $\delta + (1 - \delta)p_0 < 1$, we have $e_{high}^*(\theta) < e_U^*(\theta)$ for all θ . Recalling that $e_U^*(\bar{\theta}) = \hat{e}$, the high-regime corner $\hat{e}$ is optimal within $[\hat{e}, \bar{e}]$ for all $\theta \leq \bar{\theta}$, while for $\theta > \bar{\theta}$ the interior solution $e_{high}^*(\theta) > \hat{e}$ is feasible.*

The urban child's problem is globally concave— V_U is strictly concave in e because $w_{ee} < 0$ and $C'' > 0$ —so a unique interior solution exists. Monotonicity of e_U^* in θ follows from the supermodularity condition $w_{e\theta} > 0$: higher ability shifts the marginal-return schedule upward, raising the crossing point with $C'(e)$. The rural child cannot apply a single first-order condition because $p(e)$ is discontinuous at $\hat{e}$. The correct solution procedure is regime comparison: evaluate the best investment in each regime and choose the higher-payoff option. The over-investment result (Proposition 2) focuses on the intermediate-ability range $\theta \in (\hat{\theta}, \bar{\theta})$, where the high-regime maximum is at the corner $\hat{e}$ and $\hat{e} > e_U^*(\theta)$.

4 Under-Investment and Over-Investment

4.1 The Under-Investment Benchmark

When the rural child's optimum lies below the escape threshold, the proportional penalty delivers the [Arrow \(1973\)](#) result in the human-capital setting.

Proposition 1 (Under-Investment Benchmark). *Suppose $e_R^*(\theta) < \hat{e}$. Then $e_R^*(\theta) < e_U^*(\theta)$ for all $\theta \in [\theta_L, \theta_H]$. The investment gap $\Delta e^*(\theta) := e_U^*(\theta) - e_R^*(\theta)$ is strictly positive and strictly decreasing in δ , with $\Delta e^*(\theta) \rightarrow 0$ as $\delta \rightarrow 1$.*

The proof is a single-crossing argument. The urban child's marginal return is $w_e(e; \theta)/r$; the rural child's is $\delta w_e(e; \theta)/r < w_e/r$ at every e . Since $C'(e)$ is strictly increasing and identical across children, the intersection of marginal benefit and marginal cost occurs at a strictly lower e for the rural child. The mechanism is purely the return channel: the multiplicative penalty scales down the payoff to every unit of education, reducing investment.

Remark 1 (Scope of Proposition 1). *The under-investment result depends critically on the multiplicative penalty specification. Under an additive penalty $W_R = w(e, \theta) - d$, the rural child's marginal return is $w_e(e; \theta)/r$ —identical to the urban child's—so $e_R^*(\theta) = e_U^*(\theta)$ and the investment gap is zero. The multiplicative form is the mechanism through which the institutional penalty distorts the education investment margin.*

4.2 Escape-Route Over-Investment

The central result of this paper is that the direction of investment distortion reverses for intermediate-ability rural children. Define two ability thresholds:

$$\bar{\theta} : e_U^*(\bar{\theta}) = \hat{e}, \quad (6)$$

$$\tilde{\theta} : V_R^{\text{low}}(e_{\text{int}}^*(\tilde{\theta}); \tilde{\theta}) = V_R^{\text{high}}(\hat{e}; \tilde{\theta}). \quad (7)$$

$\bar{\theta}$ is the ability level at which the urban child's unconstrained optimum equals $\hat{e}$; for $\theta < \bar{\theta}$, the urban child would invest below $\hat{e}$ if unconstrained. $\tilde{\theta}$ is the ability level at which the rural child is indifferent between the low-regime interior solution and investing exactly at $\hat{e}$.

Proposition 2 (Escape-Route Over-Investment). *If $p_0 \geq p_0^*(\delta, \hat{e}, \bar{\theta})$ (defined in the proof), then $\tilde{\theta} < \bar{\theta}$ and*

$$e_R^*(\theta) = \hat{e} > e_U^*(\theta) \quad \text{for all } \theta \in (\tilde{\theta}, \bar{\theta}).$$

A rural child of intermediate ability invests strictly more in education than an equally-able urban child.

The proof has three steps. *Step 1.* $\bar{\theta}$ is well-defined: $e_U^*(\theta)$ is continuous and strictly increasing in θ , and by calibration $e_U^*(\theta_L) < \hat{e} < e_U^*(\theta_H)$, so the intermediate value theorem delivers a unique $\bar{\theta}$. *Step 2.* Define the payoff difference $F(\theta) := V_R^{high}(\hat{e}; \theta) - V_R^{low}(e_{int}^*(\theta); \theta)$. At $\theta = \bar{\theta}$, the escape premium is $(1 - \delta)p_0 \cdot w(\hat{e}, \bar{\theta})/r > 0$. There exists a threshold p_0^* such that for $p_0 \geq p_0^*$, this premium exceeds the net cost of over-investing from $e_{int}^*(\bar{\theta})$ to $\hat{e}$, giving $F(\bar{\theta}) > 0$. *Step 3.* $F(\theta)$ is strictly increasing in θ : by supermodularity ($w_{e\theta} > 0$) and the envelope theorem applied to the low-regime optimum, higher ability raises $w(\hat{e}, \theta)$ by more than it raises $w(e_{int}^*(\theta), \theta)$, because $\hat{e} > e_{int}^*$. Since $F(\theta_L) < 0$ (negligible escape premium at low ability) and $F(\bar{\theta}) > 0$, the intermediate value theorem delivers a unique $\tilde{\theta} \in (\theta_L, \bar{\theta})$ satisfying $F(\tilde{\theta}) = 0$. For $\theta \in (\tilde{\theta}, \bar{\theta})$, $F(\theta) > 0$ implies $e_R^*(\theta) = \hat{e}$. Since $e_U^*(\theta) < \hat{e}$ for $\theta < \bar{\theta}$, over-investment follows.

The economic mechanism is an asymmetric prize structure. Urban children gain nothing from crossing $\hat{e}$: they already earn the full urban wage at any education level. Rural children gain an expected wage jump of $(1 - \delta)p_0 \cdot w(\hat{e}, \theta)/r$ upon crossing. For intermediate-ability rural children whose unconstrained optimum lies just below $\hat{e}$, this escape premium outweighs the marginal cost of the additional investment required to reach the threshold. The result overturns the central prediction of standard discrimination models—that institutional barriers uniformly suppress investment by the affected group—by showing that a threshold-based escape route creates over-investment incentives for those close enough to the threshold to find the gamble rational.

Remark 2 (Robustness to Penalty Specification). Unlike Proposition 1, the over-investment result is robust to the penalty specification. Under an additive penalty $W_R = w - d$, the escape premium at $\hat{e}$ becomes $d \cdot p_0/r > 0$, which still creates an incentive to invest at $\hat{e}$ for intermediate-ability rural children. The over-investment direction survives; the under-investment direction does not.

5 Comparative Statics

Proposition 3 (Comparative Statics on the Investment Gap). *(i) For $\theta \in [\theta_L, \tilde{\theta}]$: $\partial \Delta e^*(\theta)/\partial \delta < 0$. Relaxing the hukou penalty reduces the under-investment gap.*

(ii) $\partial(\bar{\theta} - \tilde{\theta})/\partial p_0 > 0$. A higher escape probability expands the ability range over which rural children over-invest.

(iii) The sign of $\partial \tilde{\theta}/\partial \delta$ depends on a parameter condition. By the implicit function theorem and the envelope theorem applied to $F(\theta) := V_R^{high}(\hat{e}; \theta) - V_R^{low}(e_{int}^(\theta); \theta)$:*

$$\text{sign}\left(\frac{\partial \tilde{\theta}}{\partial \delta}\right) = -\text{sign}[(1 - p_0)\hat{e}^\beta - (e_{int}^*)^\beta].$$

When $\hat{e}/e_{int}^(\tilde{\theta}) > (1 - p_0)^{-1/\beta}$: $\partial \tilde{\theta}/\partial \delta < 0$, so a relaxation of the hukou penalty (higher δ) lowers θ and expands the over-investment range. Under the complementary condition ($\hat{e}/e_{int}^* \leq (1 - p_0)^{-1/\beta}$): $\partial \tilde{\theta}/\partial \delta \geq 0$ and stricter enforcement*

widens the range. The aggregate effect of enforcement on the over-investment range is therefore parameter-dependent and not generically monotone.

Parts (i) and (iii) follow from the implicit function theorem applied to the low-regime first-order condition and indifference condition (7), respectively. Part (ii) follows because $F(\theta)$ is strictly increasing in p_0 at every θ , which shifts $\tilde{\theta}$ downward while leaving $\bar{\theta}$ unchanged ($\bar{\theta}$ depends only on e_U^* , which is independent of p_0).

The sign result in part (iii) has a clean intuition. When $\hat{e} \gg e_{int}^*$ (first condition), a marginal increase in δ raises the high-regime base return $\delta w(\hat{e}, \theta)$ by much more than it raises the low-regime return $\delta w(e_{int}^*, \theta)$, since $\hat{e}^\beta \gg (e_{int}^*)^\beta$. Even though the escape premium $(1 - \delta)p_0 w$ falls, the net payoff advantage of the high regime increases, drawing more rural children toward over-investment. When $\hat{e}$ is close to e_{int}^* (second condition), the escape premium effect dominates and stricter enforcement widens the range. Neither direction is empirically universal; calibration to Chinese Gaokao data is required to determine which regime obtains.

6 Welfare

Let $e^{FB}(\theta) = e_U^*(\theta)$ denote the first-best investment level and define the welfare loss from hukou as $\Delta W(\theta) := V_U(e^{FB}; \theta) - V_R(e_R^*; \theta)$.

Proposition 4 (Welfare Loss). *For all $\theta \in [\theta_L, \theta_H]$ and $\delta \in (0, 1)$:*

- (i) $\Delta W(\theta) > 0$.
- (ii) $\partial \Delta W(\theta) / \partial \delta < 0$: *relaxing the penalty reduces welfare loss.*
- (iii) $\Delta W(\theta)$ *is non-monotone in θ , attaining a local maximum near $\tilde{\theta}$.*

The non-monotone profile in part (iii) reflects two competing distortions. In the under-investment region $\theta < \tilde{\theta}$, the gap between e^{FB} and e_R^* grows as θ approaches $\tilde{\theta}$ from below: $e_U^*(\theta)$ rises toward $\hat{e}$ while $e_{int}^*(\theta)$ is pushed downward by the penalty. In the over-investment region $\theta \in (\tilde{\theta}, \bar{\theta})$, the rural child invests at $\hat{e}$ and bears the cost of over-shooting with certainty, while receiving the escape premium only with probability p_0 . The welfare loss peaks near $\tilde{\theta}$ where both distortions are simultaneously active.

A natural prior predicts that welfare losses should be largest for high-ability rural children, who have the most to gain from equal treatment. Proposition 4(iii) qualifies this: above $\bar{\theta}$, rural children invest in the high regime at $e_{high}^*(\theta) > \hat{e}$, but the effective multiplier $\delta + (1 - \delta)p_0 < 1$ means their expected wage still falls short of the first-best; $\Delta W(\theta) > 0$ for all θ . What the non-monotonicity says is that the welfare loss is particularly large near $\tilde{\theta}$, where two distortions coincide: the rural child over-shoots to $\hat{e}$ (incurring excess investment cost) while still receiving the penalized wage with probability $1 - p_0$. Whether the welfare loss is monotonically smaller above $\bar{\theta}$ than near $\tilde{\theta}$ depends on parameters and is not established in general; the proposition asserts

only that $\Delta W(\theta)$ attains a local maximum near $\tilde{\theta}$, not a global ordering across the full ability distribution.

Proposition 5 (Unintended Consequence of Hukou Reform). *Let $\delta^* \in (0, 1)$ be the pre-reform penalty. A reform setting $\delta' = 1$:*

- (i) *Eliminates all welfare loss: $\Delta W(\theta; \delta') = 0$ for all θ .*
- (ii) *Increases investment for $\theta \in [\theta_L, \tilde{\theta}(\delta^*)]$: $e_R^*(\theta; \delta') = e_U^*(\theta) > e_R^*(\theta; \delta^*)$.*
- (iii) *Decreases investment for $\theta \in (\tilde{\theta}(\delta^*), \bar{\theta})$: $e_R^*(\theta; \delta') = e_U^*(\theta) < \hat{e} = e_R^*(\theta; \delta^*)$.*

The proof is direct. Post-reform ($\delta = 1$), both children solve the same problem and arrive at the first-best: $e_R^*(\theta; 1) = e_U^*(\theta)$. Welfare is equalized by construction. The investment comparison in (iii) follows because the pre-reform rural child in $(\tilde{\theta}, \bar{\theta})$ was investing at $\hat{e} > e_U^*(\theta)$, while the post-reform rural child invests at $e_U^*(\theta)$.

The paradox arises because the escape-route over-investment was a distortion, not genuine educational ambition. Once the wage penalty disappears, intermediate-ability rural children optimally reduce education to the unconstrained level: they earn the full wage at lower cost, strictly improving welfare. A policymaker who interprets reduced Gaokao preparation intensity among rural near-threshold students as evidence that reform has backfired is drawing the opposite conclusion from the correct one: the observed investment decline is the efficiency gain from removing the distortion.

7 Boundary Cases

Proposition 6 (Limiting Behavior). (i) *As $\delta \rightarrow 1$: $e_R^*(\theta) \rightarrow e_U^*(\theta)$ and $\Delta W(\theta) \rightarrow 0$ for all θ .*

(ii) *As $p_0 \rightarrow 0$: the over-investment range collapses, $\bar{\theta} - \tilde{\theta} \rightarrow 0$. For $\theta \leq \bar{\theta}$, $e_R^*(\theta) \rightarrow e_{int}^*(\theta)$ (the low-regime interior). For $\theta > \bar{\theta}$, the high-regime interior $e_{high}^*(\theta) \rightarrow e_{int}^*(\theta)$ as well (since the effective multiplier $\delta + (1 - \delta)p_0 \rightarrow \delta$), representing a contraction of investment for these high-ability children. The overall limit is pure under-investment relative to first-best for all θ , consistent with the [Arrow \(1973\)](#) benchmark.*

(iii) *As $\delta \rightarrow 0$: $e_{int}^*(\theta) \rightarrow 0$ for all θ ; the escape route becomes the only investment incentive for rural children. Whether rural children actually cross $\hat{e}$ depends on whether the escape premium $p_0 w(\hat{e}, \theta)/r$ exceeds $C(\hat{e})$; for very low δ and low θ , neither regime may generate positive investment.*

The limit in (ii) is the most informative. As $p_0 \rightarrow 0$, the payoff jump at $\hat{e}$ vanishes; the rural child always prefers the low-regime interior, yielding the Arrow (1973) prediction of proportional under-investment at every ability level. For high-ability children ($\theta > \bar{\theta}$),

convergence to the Arrow benchmark involves a fall in investment from the high-regime interior $e_{high}^* > \hat{e}$ back to $e_{int}^* < \hat{e}$, not merely a collapse of the intermediate over-investment range. The novel predictions of the present paper—escape-route over-investment (Proposition 2) and the reform paradox (Proposition 5)—require both $\delta < 1$ and $p_0 > 0$; neither feature alone is sufficient.

8 Testable Predictions

The model generates the following empirically testable predictions.

P1. Non-monotone investment gap. Among rural and urban students of equal measured ability, the investment gap $\Delta e^*(\theta)$ is positive (rural under-invest) at low ability, negative (rural over-invest) at intermediate ability near the Gaokao key-line, and approaches zero at high ability. Cross-sectional variation in Gaokao preparation intensity by ability and hukou status provides the most direct test.

P2. Over-investment expands under stricter enforcement. When the elite threshold substantially exceeds rural students’ unconstrained optimum (Proposition 3(iii)), stricter regional hukou enforcement should increase preparation intensity among near-threshold rural students. Regional variation in enforcement strictness is the natural testing ground.

P3. Post-reform investment decline among near-threshold students. In regions where hukou reform has reduced the wage penalty, Gaokao preparation intensity among intermediate-ability rural students should fall even as wage and welfare outcomes improve. Isolating this prediction from general discouragement effects requires joint measurement of wage outcomes and investment.

P4. Welfare loss concentrated at intermediate ability. Welfare measures—earnings gaps, consumption shortfalls, subjective well-being differentials relative to urban peers—should be largest for rural students near the elite admission threshold, not for the most- or least-able.

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